System Description of ShanHe-Huawei_OceanStor_A800_8host_1A800

1. Basic Information

1.1 System name of the submission

ShanHe-Huawei_OceanStor_A800_8host-1A800

1.2 Submitter name

JNIST

1.3 Submission date

July-10-2025

1.4 Version of the benchmark

MLPerf Storage V2.0

1.5 Solution type of the submission

Proprietary - OceanStorOS

1.6 Submission division (OPEN or CLOSED)

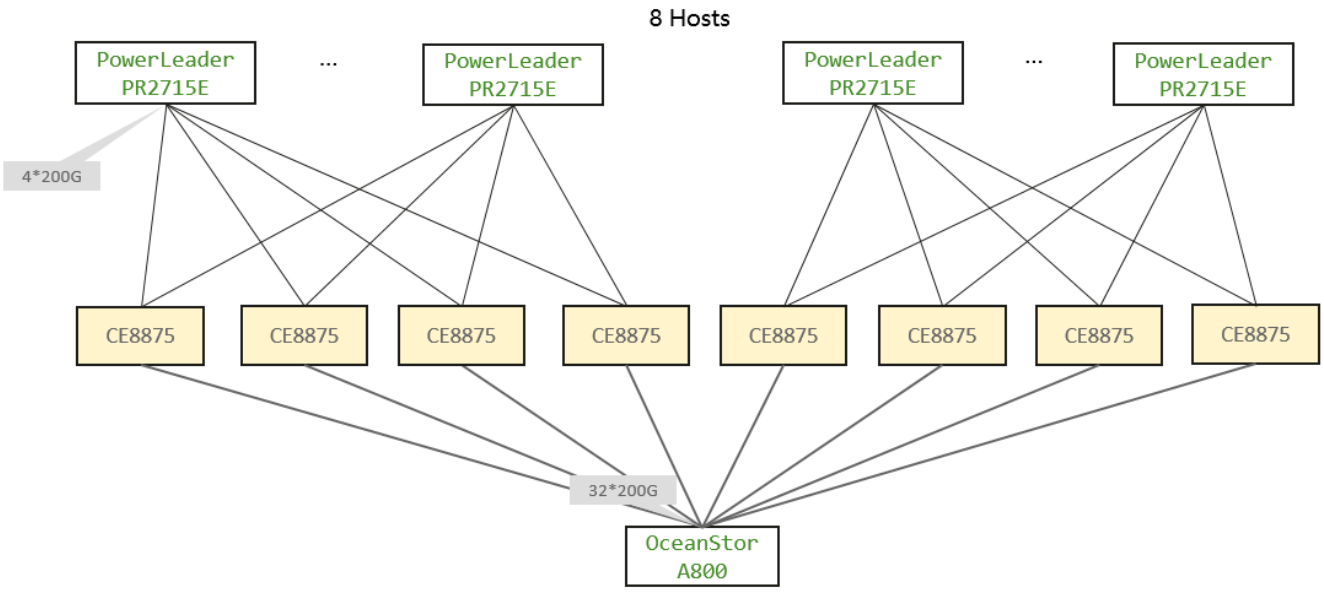
CLOSED

1.7 Power Requirements

System component	Power supply unit	Nameplate rated power	Design Power
A800 Appliance 1	Power supply 1	2000W	6000W
	Power supply 2	2000W	
	Power supply 3	2000W	
	Power supply 4	2000W	
	Power supply 5	2000W	
	Power supply 6	2000W	
	Power supply 7	2000W	
	Power supply 8	2000W	
Totals		16000W	6000W

1.8 System Topology

The test environment consists of 8 hosts and one Huawei OceanStor A800, connected to eight ethernet RoCE switches.



1.9 Rack Units Requirements

The Storage system consumes 8U space.

2. Hardware

2.1 Host

The host nodes are commercial servers with 2 Intel(R) Xeon(R) Platinum 8458P Processors. The Clients communicate with the storage through RoCE. The hardware details are as follows:

Component	Description	Count
CPU	Intel(R) Xeon(R) Platinum 8458P	2
Memory	DDR5 4800MHz 32GB	16
Network Card	Mellanox Technologies MT2892 Family [ConnectX-6 Dx]	4

2.2 Storage

The Huawei OceanStor A800 is a new generation of high-performance AI storage specifically designed for AI. It adopts an innovative control-data separation architecture, delivering ultimate performance to accelerate the training and inference of large models.
The Huawei OceanStor A800 high-performance AI storage supports flexible scaling, with a single cluster capable of supporting 512 controllers for scale-out and EB-level storage capacity expansion, meeting the smooth evolution needs of trillion/quadrillion-parameter multimodal large models. Additionally, the storage natively supports new data paradigms such as vectors, with a built-in high-performance vector retrieval engine that accelerates the retrieval of large-capacity vector knowledge bases, reduces inference "hallucinations," and provides KV Cache multi-level caching capabilities to reduce inference latency and improve the cost-effectiveness of inference. This storage system meets the full-process data processing needs of AI training and inference.



The hardware details of one appliance are as follows:

Component	Description	Count
CPU	HiSilicon Kunpeng-920	8
Memory	DDR5 32GB	64
Network Card	2*200G RoCE	16
Drive	NVMe SSD 7.68TB	64

2.3 Switch

Eight Huawei CloudEngine 8875 switches are used. The host and storage are connected to it.

3 Software

3.1 Host

The configuration of the host software is shown as follows:

Software	Description
OS	openEuler release 22.03 (LTS-SP4)
Kernel	5.10.0-216.0.0.115.oe2203sp4.x86_64
Mellanox OFED version	MLNX_OFED_LINUX-5.8-1.1.2.1

3.2 Storage

The software deployed on the storage is OceanStor A800 V700R001C10.